

RAHUL DHIMAN

Mine Technology & AHS Readiness Lead | Fleet Management Systems (FMS) & Dispatch (Wenco), Mine Control, Operational Technology (OT), PLC/SCADA
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PROFESSIONAL SUMMARY

Mining technology and operations professional with experience since 2016 in open-pit mining and mineral processing in Indonesia and the UAE. Wenco Dynamic Dispatch certified; led a greenfield FMS implementation from design to go-live, supporting a rise in daily production from 60,000 to 90,000 tonnes. Now works on dispatch, mine control and safety systems at a multi-site nickel operation with more than 1,000 machines. Leads AHS readiness work on the network, workflows and data needed before autonomy can run. Builds practical decision tools: a data-driven production simulator, QR identity and permit-verification workflows, a dispatch concept prototype and a live-data safety dashboard prototype. Lean Six Sigma Black Belt; pursuing a Doctorate in Management focused on AI and autonomous systems in mining.

CORE COMPETENCIES

Fleet & Dispatch: fleet management systems (Wenco FMS, Dynamic Dispatch certified), dispatch and safety platforms, mine control room operations, dispatch configuration, cycle-time and queue optimisation, payload management, AHS readiness

Mine Safety Technology: AI driver monitoring (fatigue, seatbelt), ADAS forward collision warning, geofencing and automated speed enforcement, 24/7 control room design, ISO 45001, EN ISO 13849-1, AS/NZS 4871 lockout-tagout

Automation & Operational Technology (OT): Siemens S7-1500/1200/300 (TIA Portal V15-V17), Allen-Bradley ControlLogix (Studio 5000), Structured Text, Function Block, Ladder, SCADA/HMI (WinCC, FactoryTalk View SE), PID tuning, VFD commissioning, OT/IT integration

Networks & Positioning: Rajant Kinetic Mesh, industrial wireless, VHF/UHF and 4G site communications, fibre backbones with RSTP, VLAN and QoS, Ethernet/IP, Modbus TCP, Trimble GNSS base stations

Data & Improvement: Power BI with DAX, SQL, Python automation, OSIsoft PI, SAP PM integration, maintenance analytics, continuous improvement, DMAIC, root cause analysis, Lean Six Sigma Black Belt, drone survey and photogrammetry

Digital Builds & Delivery: in-house tool development with Python (Flask, FastAPI), SQL Server and React — production simulation, QR identity and permit verification, dashboard and FMS prototyping; stakeholder coordination with contractors and vendors; implementation and commissioning

PROFESSIONAL EXPERIENCE

PT ERAMET INDONESIA — WEDA BAY NICKEL PROJECT | Weda, Indonesia

Fleet Management System Engineer | Oct 2024 – Present

Responsible for integrated dispatch and safety technology at a multi-site nickel operation with more than 1,000 machines.

- Led dispatch and safety platform deployment using real-time GPS, speed and onboard payload telemetry, helping reduce dispatch response time from 15–20 minutes to 2–3 minutes.
- Stood up 24/7 mine control rooms with multi-monitor dispatch consoles and haul-road and stockpile CCTV, and extended VHF/UHF and 4G coverage with IT and OT teams across the haul-road network, sustaining 95%+ platform uptime.
- Configured geofencing and automated speed enforcement, contributing to around 40% fewer speeding violations, and commissioned vendor-provided AI driver monitoring and ADAS forward collision warning.

- Leads Autonomous Haulage System (AHS) readiness work on the network, workflows and data needed before autonomy can run.
- Led digital radio network rollout and operator-ID linkage, supporting reliable site communications and emergency response.
- Built Power BI dashboards with DAX to monitor mileage, tonnage, fuel and service intervals for maintenance planning.
- Worked on drone-based stockpile volume measurement and survey workflows.
- Coordinated onboard hardware installation and trained more than 3,000 operators, with technical-staff support, documentation and refresher programmes.
- Built a production simulator (Python, SQL Server) using measured haul history to estimate trip times, trips per shift, tonnes per route and fleet requirements, with scenario comparison for planning. Developed and tested as an internal tool and demonstration platform.
- Built a QR identity and permit-verification system linking worker credentials, training, assigned vehicles and inspection workflows.
- Built an in-house FMS concept prototype (React, TypeScript) with dispatch, loading and dump-management views, using mock data and offline snapshots as a reference for vendor discussions and trials.
- Built a read-only safety dashboard prototype connected to fleet telematics (FastAPI, ECharts), covering overspeed intensity, intervention effectiveness, risk mix and data quality.
- Audited the model against measured shift data, found a cycle-time definition error that overstated forecast production, and corrected it to measured effective cycle time per route.

STEVIN ROCK LLC | Ras Al Khaimah, UAE

Production Control & Automation Lead Engineer | Feb 2024 – Sep 2024

Led reliability and safety improvement across PLC-automated crushing operations at a high-volume limestone and aggregates site.

- Cut unplanned downtime by around 50% through real-time monitoring of the crushing lines, and more than halved weekly emergency stops with automatic feed-rate logic that prevents material bridging.
- Led installation of a camera monitoring network across the crusher, conveyors, tipping locations and other critical points, improving remote operational visibility.
- Designed, established and commissioned the crusher control room, enabling operators to control the entire crushing circuit remotely.
- Relocated operators out of high-noise, high-dust field positions into the new control room, with video walls and dual-monitor SCADA consoles, cutting noise-exposure complaints by around 60%.
- Built a continuous workflow between the crushing plant and the quarry, working with the quarry FMS team to synchronise truck movements with crusher demand, reducing downtime on both sides.
- Aligned circuit throughput to the period's production requirement, setting and monitoring the TPH target and supervising the operator team against it.

STEVIN ROCK LLC | Ras Al Khaimah, UAE

Fleet Management System (FMS) Shift Engineer | Jan 2022 – Feb 2024

Appointed as the site's first FMS engineer, leading the project from tower survey and mesh network build through Wenco commissioning and live dispatch optimisation at a site with no fleet management system.

- Supported a rise in daily production from 60,000 to 90,000 tonnes through dispatch optimisation, onboard payload monitoring and structured operator coaching.
- Led the network build: surveyed tower positions and commissioned solar-powered Rajant towers, radios, repeaters and links site-wide, sustaining a 24-hour mesh network in terrain where grid power could not reach.
- Assisted Wenco engineers with fleet hardware installation (onboard computers, GNSS, radios, payload systems) and completed Wenco Dynamic Dispatch training before go-live.
- Implemented dynamic dispatch: tuned queue, loading and dumping rules shift by shift, and coordinated loading and hauling equipment, holding excavator queues under two minutes.

- Tracked productivity of every loading and hauling fleet — waiting time, operating time, tonnes per hour and efficiency per unit by shovel and excavator, and payload variance.
- Extended the platform into fuel, brake and shift-start management, and integrated predictive maintenance alerts with SAP PM for automatic work orders.
- Moved FMS data into SQL and built Power BI dashboards reviewed hourly with site supervisors to revise the plan within the same shift.

STEVIN ROCK LLC | Ras Al Khaimah, UAE

Automation & Production Engineer / Electrical Engineer | Jan 2016 – Jan 2022

Led automation and daily crushing production across a large crushing complex, coordinating plant throughput and the electrical, PLC and SCADA systems behind it.

- Led delivery of Structured Text and Function Block logic (TIA Portal) for conveyors, crushers and screening plants, sustaining availability above 96%.
- Programmed Allen-Bradley ControlLogix, safety interlocks and PID feed control, holding throughput within +/-5% of target.
- Ran daily crusher production and throughput optimisation — feed rate, crusher settings, screen selection — clearing bottlenecks shift by shift with the electrical and maintenance teams to hold the production target.
- Planned daily and weekly plant production, sizing product to customer specification to get the maximum saleable output from every tonne of feed.
- Commissioned variable frequency drives and soft starters (ABB and Siemens) with optimised ramp, current-limit and energy-saving profiles; regenerative braking on decline conveyors helped cut electrical cost by around 10%.
- Sustained electrical availability of around 97% through planned maintenance of motor starters, substations and PLC cabinets.
- Built WinCC and FactoryTalk View interfaces with live KPIs and alarm escalation over resilient industrial networks.
- Integrated Metso optical rock-sizing technology over Modbus TCP to support process control and product sizing.

EDUCATION

Doctorate in Management, research focus: AI & Autonomous Systems in Mining | UCAM, Spain | In progress

M-Tech, Manufacturing Management | BITS Pilani (WILP), India | 2024

University Diploma, Mineral Extractives Studies | University of Derby, United Kingdom | 2024

MBA, Operations & Management | Kalinga University, India | 2015

B-Tech, Electronics & Communication Engineering | Punjab Technical University, India | 2013

CERTIFICATIONS

Wenco Fleet Management System — Dynamic Dispatch | Wenco International Mining Systems | 2022

Project Management Professional (PMP) training — 35 contact hours | 2024

Emergency First Aid, Defibrillation and CPR | Highfield | 2024

Lean Six Sigma Black Belt | CPD Certified | 2023

Microsoft Power BI Data Analyst Professional Certificate | 2023

McKinsey Forward Program | McKinsey & Company | 2023

PLC Automation, Siemens TIA Portal | 2021

Diploma in Quarry Management | Institute of Quarrying | 2018

LANGUAGES

English (fluent) | Hindi (native) | Bahasa Indonesia (conversational) | French (basic)

ADDITIONAL INFORMATION

Target roles: Mining Technology Manager | Mine Control Superintendent | Fleet Management System (FMS) Lead / Specialist | AHS Readiness Lead | Product Manager (mining technology) | Operational Technology.

Based in Indonesia; open to international relocation.

Worked across the UAE and Indonesia, coordinating multilingual contractor teams. Member, Project Management Institute (PMI).